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Graphene Supported Boron Nitride Nanosheets as Advanced Electroanalytical Performance for Rechargeable Magnesium Storage System

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Graphene supported boron nitride (G/BN) nanosheets were prepared via vacuum filtration process and followed by thermal treatment at 150 °C for 2 h in an argon atmosphere. The structural analysis of G/BN materials were conducted through field emission scanning electron microscopy, energy-dispersive X-ray spectroscopy, X-ray diffraction, X-ray photoelectron spectroscopy, FT-IR spectroscopy. The structural analysis was clearly illustrated that the prepared BN nanosheets were supported on the graphene layer. It is confirmed that the G/BN-20% content shown round leaf-like morphology and

particle size approximately 100 nm. Further, the electroanalytical performance of G/BN materials was measured by galvanostatic charge/discharge, cyclic voltammetry, and electrochemical impedance spectroscopy studies. It was observed that the G/BN-20% material was exhibited with a good discharge capacity (105 mAh g⁻¹) at the current density of 25 mA g⁻¹, which is higher capacity retention and rate capability at around 100 cycles. Due to the exceptional magnesium storage behaviour of G/BN-20% material which is used as a cathode, has shown a strong synergistic effect.

Introduction

Nowadays, energy storage systems are more important in the modern world. Commercial rechargeable batteries like Li-ion batteries (LIBs) are the great powerful scientific properties for portable electronic devices. Most electric and hybrid vehicles currently used LIBs as an energy source due to their extraordinary power and energy density. Research in topical years has shown that the LIBs have many drawbacks, such as relatively low abundances, high costs, and safety limitations. To overcome the above issues, magnesium battery is the most suitable storage system which is associated with LIBs since, the magnesium battery has low a material cost and safety.^[1–3] Metallic magnesium is an anode material that possesses a high theoretical capacity (2250 mAh g⁻¹) and negative electrode potential (-2.37 V). Reduction in the total cost of metallic magnesium anode material is due to its natural abundance.^[4] However, it is highly difficult for Mg insertion/extraction in the

cathode host structure due to a high polarization of divalent Mg ions.^[5]

The work was mainly focussed on a particular electrolyte because of nonconducting passivation film to form on the Mg surface of the electrode in organic electrolyte.^[6] Many previous studies have shown the electrolyte, which is non-aqueous Grignard organoboranes in ethereal solvents was suitable for Mg batteries, although, these ethereal solvents are the main drawback of voltage restrictions.^[7] It is necessary to develop a proper electrolyte for the future magnesium battery application. Therefore, imidazolium-related electrolytes were displayed an extensive range of applications such as good conductivity, non-flammability, great thermal stability, good electrochemical stability, lower vapor pressure, and extensive electrochemical window.^[8–9] So, 1,3 dibutyl imidazolium bromide ([bim][Br]) has a superior behaviour in the rechargeable batteries.

Boron nitride (BN) nanosheet has a graphene-like structure in which N and B atoms are substituted in a position of C atoms,^[10–11] which can be generally called as graphene. In general, BN exhibit as insulating material that could not show electrochemical performance. The selective doping process will decrease the band gap of BN and enhance the good electrical property. Now, BN is used in many different areas because of high mechanical properties, high chemical stability, and good thermal conductivity.^[12–13] The monolayer of graphene has sp² carbon atoms and arranged in the two-dimensional form of hexagonal lattice^[14–16] which exhibits a wide range of electrical properties due to its remarkable structure flexibility, superior electrical conductivity, high surface area and high chemical tolerance.^[17–22] Layered transition metal sulfide (such as WS₂ and MoS₂)/graphene composites are generally used for Li-ion batteries with good electrochemical performances as a result of its synergistic effects in between graphene and transition metal

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sulfide.^[23–24] In recent, boron nitride/graphene oxide composite used in the asymmetric supercapacitor devices^[25] since these electrodes showed a significant improvement in electrochemical performances. In view of the above examples, we designed both graphene and boron nitride nanosheets (G/BN) as a cathode material for magnesium batteries. It is expected to deliver the intrinsic for more efficient charge carrier and storage for magnesium batteries.

In this work, a simple method is adopted to prepare graphene/boron nitride composite with different BN ratios by using a vacuum filtration technique followed by thermal treatment. Composite based graphene with a minimum amount of BN substance will exhibit good electrochemical stability with high reversible capacity, good rate capability, good cyclic performance, and good coulombic efficiency. Use of these materials, led to a great potential cathode for magnesium batteries.

Results and Discussion

FE-SEM & TEM analysis

Figure 1 demonstrates the morphology of h-BN, h-BNNSs, GNS and G/BN-20% composites and this morphology was scan by FESEM. The h-BN contains bulky size sheets and abnormal shapes and their sizes at around 5 μm (Figure 1(a)). The FESEM image of h-BNNSs was synthesized from h-BN powder in IPA as a solvent which denoted (Figure 1(b)), which consists of many

sheets in which each sheet was combined with each other and the shape of each sheet-like blooming flowers. The size of flakes length approximately at around 3 μm and flakes thickness approximate around 100 nm. It is clearly shown that the size and thickness of h-BNNSs was reduced. It was observed that the successive formation of BN nanosheets from h-BN powder. The graphene nanosheets (GNS) consist of mono or few-layered morphology (Figure 1(c)). The size of sheet diameter around 5 μm and sheet thickness was around 100 nm. It indicated the successive formation of graphene nanosheets from natural graphite by using an electrochemical method. The FESEM image of G/BN-20% was prepared by using vacuum filtration followed by thermal treatment at 150 °C for 2 h (Figure 1(d)). G/BN composite film shape was appeared as a round leaf-like structure. The BN nanosheets were supported on the surface of the graphene sheet (Figure 1(d)). Finally, the formation of G/BN composite has occurred and the size of sheet edge thickness is about 100 nm, respectively.

Figure 1(e)&(f) refer to the high-resolution TEM micrograph of graphene sheet (GNS) and G/BN-20% composite. In Figure 1(e), the graphene sheet has displayed the presence of single or few layers, folded or flexible and crumpled sheets. So, considering the graphene sheets which exhibit a nanometre of edge thickness. In Figure 1(f), the morphology of G/BN-20% was displayed the 2D graphene nanosheet and BN particles were well conserve after their formation of composition.^[28] From the TEM image, the approximate thickness of G/BN-20% composite was at around 0.1 μm . In addition, the morphology of the G/BN composite with different BN substances showed close to the G/BN-20% composite. Overall, the TEM results revealed that the graphene sheet (GNS) and G/BN-20% composite film have successfully formed.

EDX analysis

The EDX spectrum was elaborated on an approximate weight percentage alignment of G/BN-20% composite. It is clearly demonstrating that the major signals are B, C, and N (Figure 2). The approximate wt.% of B, C, N, and O for G/BN composite were 25.7%, 35.2%, 35%, and 4.1% respectively.

XRD analysis

XRD patterns of natural graphite, graphene nanosheet (GNS) and graphene/boron nitride (G/BN-20%) composite materials were carried out at a scan rate of 2° min⁻¹ (2 degree per minute) (Figure 3). The position of diffraction peaks was assigned as a hexagonal phase of graphite, graphene and boron nitride (JCPDS No.78-0484). XRD is a very important technique to find out the d-space of corresponding peaks. Table 1 illustrates the d-space values calculated from the XRD data of natural graphite, pure GNS, and G/BN-20% composite.

Figure 3, the graphite gave an intensive peak at 26.1° and graphene at 25° which effected its (002) crystal plane that could lead to increase in d-spacing from 3.41 to 3.56 Å,^[26] further the formation of a monolayer of graphene nanosheets have occurred. In Figure 3, G/BN-20% composite contain an

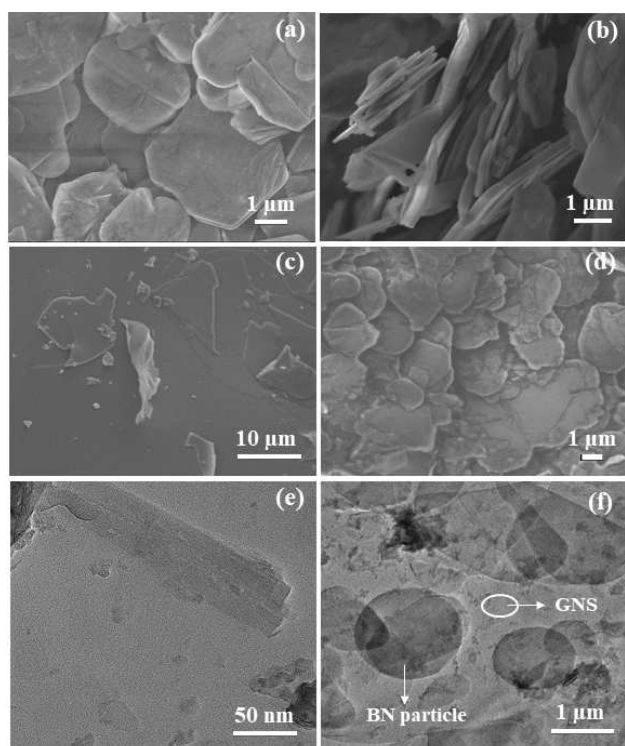


Figure 1. FESEM image of (a) h-BN, (b) h-BNNSs, (c) pure GNS, and (d) G/BN-20% composite. The TEM image of (e) pure GNS and (f) G/BN-20% composite.

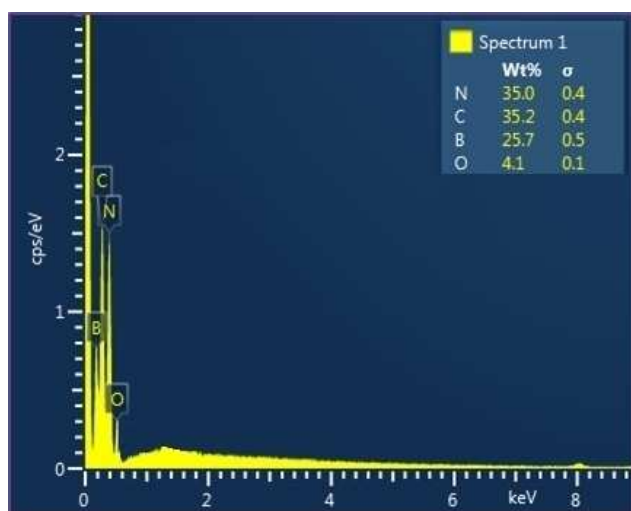


Figure 2. EDX spectrum of G/BN-20% composite.

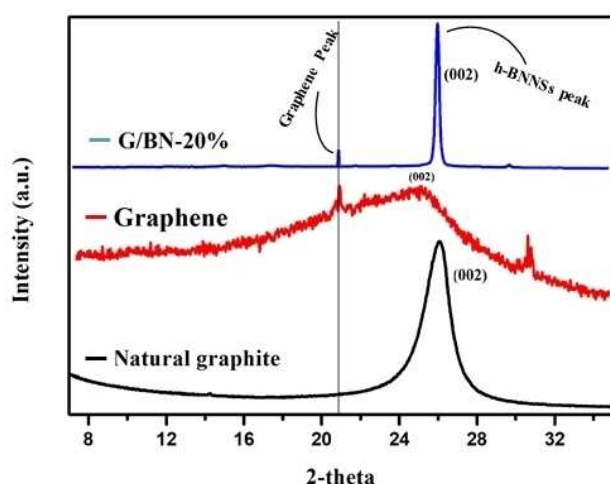


Figure 3. XRD of natural graphite, graphene nanosheet (GNS) and G/BN-20% composite.

Table 1. The d-space values of natural graphite, pure GNS, and G/BN-20% composite.		
Samples	2 θ (degrees)	d-space (Å)
Natural graphite	26.1	3.41
	25	3.56
G/BN-20% composite	21.2	4.18
	21.2	4.18
	26.7	3.334

intensive peak that occurred at 26.7° which corresponds to (002) lattice plane of h-BNNS and the sharp peak was observed at 21.2° which was similar to that of graphene peak.^[27–29] It was concluded that the formation of the G/BN composite.

XPS and FT-IR analysis

XPS analysis was carried out to identify the binding energy of the G/BN-20% composite. A wide XPS spectrum of G/BN-20% composite material consists of B 1s (188.6 eV), C 1s (282.8 eV), N 1s (396.1 eV), and O 1s (530.3 eV) (Figure 4(a)). In Figure 4(b), it is clearly observed that the C–C peak at 282.8 eV which epitomized the successful preparation of graphene from natural graphite. The N 1s spectrum in Figure 4(c), exhibits two corresponding peaks at 396.1 eV (N–B), and 397.8 eV (N–C) can be assigned as nitrogen atom binding with boron and carbon atoms respectively. XPS analysis has clearly shown that the G/BN composite was prepared from graphene and BN nanosheets.

Figure 4(d) shows the comparative FT-IR spectrum of pure graphene nanosheet (GNS), boron nitride nanosheets (BNNSs), and G/BN-20% composite. The FT-IR spectrum of pure GNS was observed with a medium peak at 1600 cm^{-1} for the C=C bond.^[30] The BNNSs sample shows a broad peak at 1375 cm^{-1} and a sharp peak at 815 cm^{-1} which indicating the stretching and bending B–N modes.^[31] In G/BN-20% composite sample, the C=C peak is almost absent (Figure 4(d)). A new peak was observed at 1650 cm^{-1} in G/BN-20% composite is due to C=N bond formation.^{[25][32]} The strong peak at 1375 cm^{-1} and a sharp peak at 815 cm^{-1} were represented to stretching and bending B–N nodes in the G/BN-20% composite. All the above results were reflected the formation of G/BN-20% composite from graphene and BN.

Cyclic voltammetry (CV) analysis

Figure 5(a) denotes the reversible study of magnesium dissolution/deposition behaviour in 0.5 M $\text{Mg}(\text{ClO}_4)_2/[\text{bim}][\text{Br}]$ electrolyte with Pt strip as a working electrode and two different Mg plates as counter & reference electrode. Use of ionic liquid electrolyte (0.5 M $\text{Mg}(\text{ClO}_4)_2$) will give good oxidation stability (Mg dissolution process) at around 1.12 V, lower negative potential for Mg deposition process and greater capacity for the reversible Mg dissolution and deposition processes.

The cyclic voltammetry was carried out on pure GNS and G/BN composite electrodes acts as working electrode with Mg electrode as counter and reference electrode in ionic liquid electrolyte at a scan rate of 5 mV/s in the voltage range from -0.8 to 1.9 V within first three cycles (Figure 5(b–e)). Four samples (GNS, G/BN-20%, G/BN-30%, and G/BN-40% composites) were exhibited different current responses at the potential window in the range of -0.8 to 1.9 V. Figure 5(b) represents the pure GNS sample in which single oxidation and reduction peak was occurred at around 1.35 and 0.85 V. Obtained two peaks were indicating to Mg-ions insertion and extraction on the graphene layer.^[33] In Figure 5(c–e), the anodic process, all G/BN composites were appeared as a single oxidation peak at around 0.86, 0.52 and 0.45 V respectively. Strong and intense peaks were represented to Mg-ion intercalation with the composite functional groups on the edge sites.^[34–37] In Figure 5(c–e), the cathodic process, single reduc-

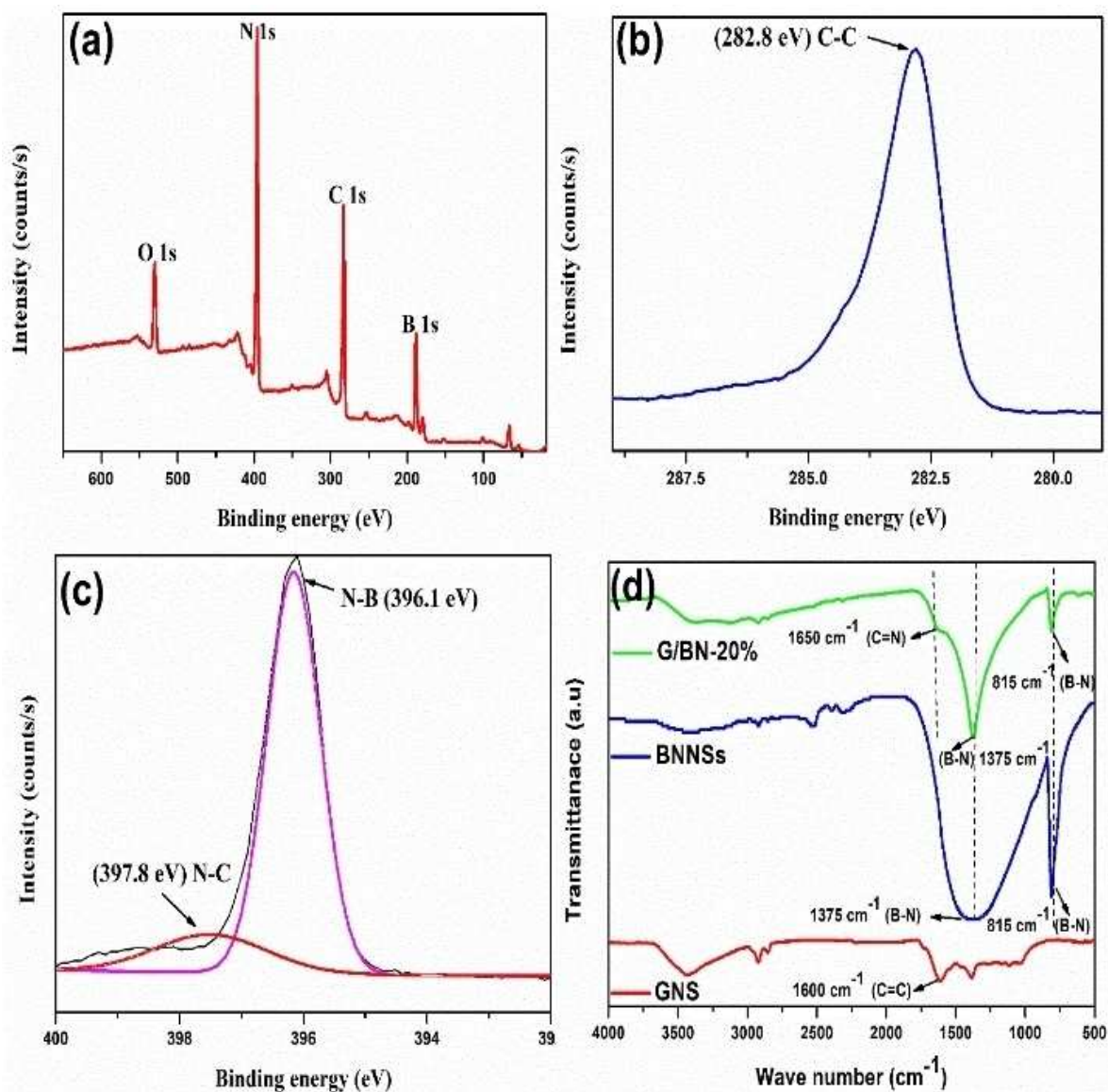


Figure 4. XPS spectrum of G/BN-20% composite (a) with a wide scan (b) C 1s scan, (c) N 1s scan and (d) FT-IR spectrum of pure GNS, BNNs, and G/BN-20% composite.

tion peaks of all G/BN composites were appeared at around 0.45, −0.25 and 0.28 V respectively. The peaks were represented the de-intercalation of Mg-ions in between the surface of composite functional groups. The current density of pure GNS and G/BN composites were calculated by using CV and found at 75 mA g^{−1}, 105 mA g^{−1}, 60 mA g^{−1}, and 50 mA g^{−1} respectively.

The peak current was gradually decreased with increasing the cycles because of the SEI formation on the surface of the electrode. The CV result revealed that the G/BN composite with 20% BN substance has good reversibility and high current density when compare to pure GNS and other composite materials since proper wt.% of graphene (G) and BN nanosheets in favourable to enable a synergistic effect. If BN

substance increases, then it will have irreversibility and lower current density because of the following main reasons (a) semiconducting property of BN: (b) proper wt.% of G and BN nanosheets does not enable synergistic structure.

Galvanostatic charge-discharge studies

Electrochemical performance of pure GNS, G/BN-20%, G/BN-30%, and G/BN-40% composite electrodes for Mg battery were further studied by charge-discharge in between 0.0 to 2.6 V. Figure 6 shows the first four-cycle charge/discharge plots of the pure GNS, G/BN-20%, G/BN-30% and G/BN-40% composites at the current density of 25 mA/g. The detailed discharge profile and rate performance of all samples are listed in Table 2.

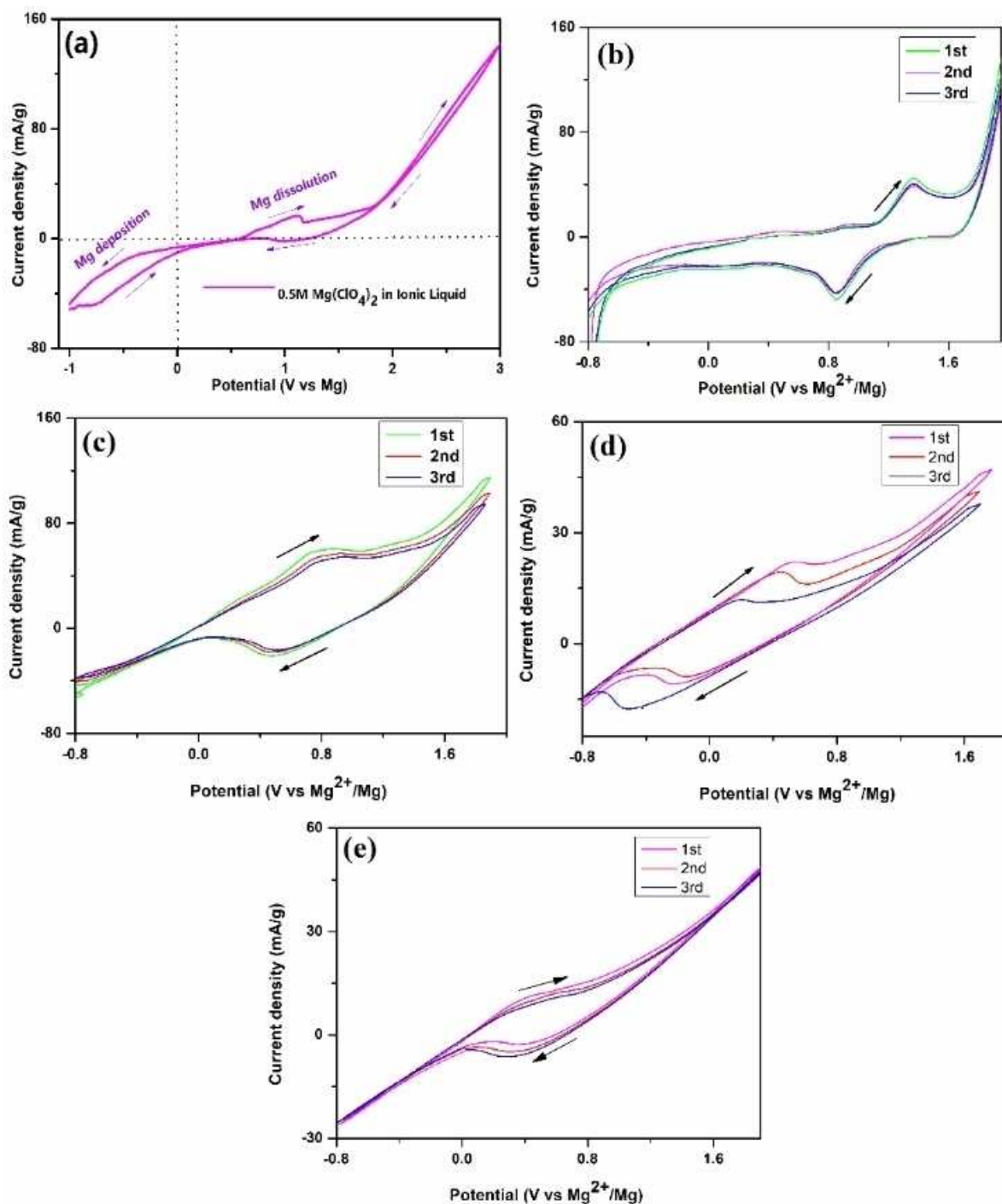


Figure 5. (a) CV of the reversible Mg dissolution/deposition using 0.5 M $\text{Mg}(\text{ClO}_4)_2$ in an ionic liquid electrolyte (Pt strip as WE and Mg as CE & RE) at 5 mVs^{-1} scan rate. Cyclic voltammetry (CV) curves of G/BN composite at scan rate of 5 mVs^{-1} in the potential window of -0.8 to 1.9 V : (b) pure GNS, (c) G/BN-20%, composite (d) G/BN-30% composite, and (e) G/BN-40% composite.

In Figure 6(a), the voltage plateau was observed at 1.1 V due to the pure GNS sample was in the fourth discharge cycles which led the Mg-ions insertion on the $\text{sp}^2\text{ C}$ atoms of the graphene layer. Figure 6(b) demonstrate that the voltage plateau at 1.0 V was observed for the G/BN-20% composite electrode in the first discharge cycle. The discharge plateau at

1.0 V represents the Mg-ion binding with the composite functional groups on the surface.^[38] Figure 6(c) & 6(d) indicate the G/BN-30% composite furthermore, it was observed voltage plateau at 0.75 V in the first discharge cycle and the G/BN-40% composite observed voltage plateau at 0.65 V in the first discharge cycle. The voltage plateau represents the Mg-ion

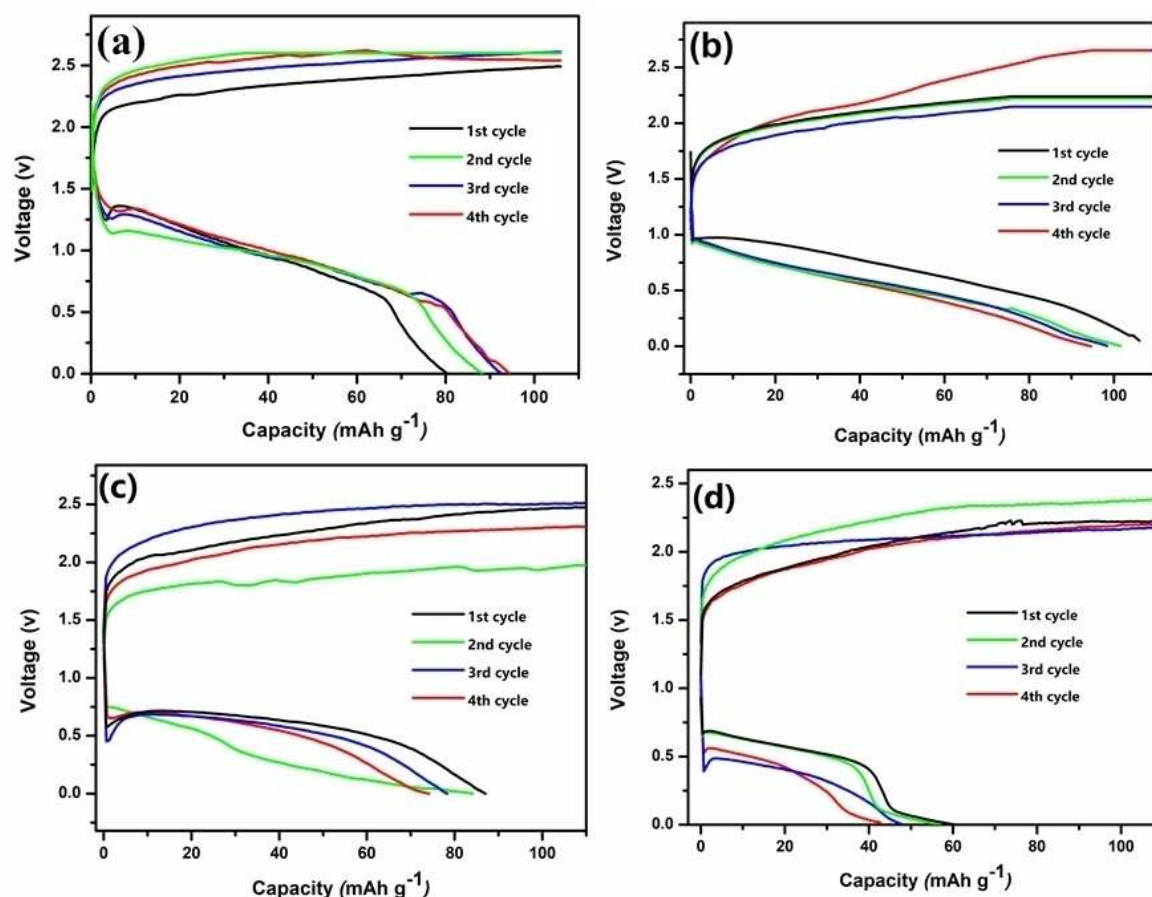


Figure 6. Galvanostatic charge-discharge profile of four samples: (a) pure GNS, (b) G/BN-20%, (c) G/BN-30% and (d) G/BN-40% composite.

Table 2. The detailed discharge profile and rate performance of four samples.

Sample	Discharge plateau (V)	Capacity mAh/g	Columbic efficiency
Pure GNS	1.1	94	90%
G/BN-20%	1.0	105	95%
G/BN-30%	0.75	85	77%
G/BN-40%	0.65	60	55%

binding with the composite functional groups. In Figure 6(b-d), the discharge voltage plateau has decreased since the formation of SEI film on the composite functional group. The discharge capacity of pure GNS and G/BN composite with different wt.% of BN substance were 94 mAhg⁻¹, 105 mAhg⁻¹, 85 mAhg⁻¹, and 60 mAhg⁻¹ respectively. The charge/discharge results were revealed that the G/BN composite with 20% BN substance has good reversibility and high discharge capacity than that of pure GNS as well as other composite materials due to proper wt.% of graphene (G) and BN nanosheets are favourable to enable a synergistic effect. The columbic efficiency of pure GNS, G/BN-20%, G/BN-30%, and G/BN-40%

composites during the charge/discharge cycle are 90%, 95%, 77% and 55% respectively.

Electrochemical impedance spectroscopy and cycle performance

Electrochemical impedance spectroscopy (EIS) was carried out after 15 cycles to understand the further electrochemical behaviour of pure GNS, G/BN-20%, G/BN-30% and G/BN-40% composites in the ionic liquid electrolyte. The Nyquist plots were recorded in the frequency range starting from 1000 kHz to 10 mHz as shown in Figure 7(a). Nyquist plots of all samples were displayed at the high-frequency region to the low-frequency region. In general, the Z_{real} axis with an arc at the high-frequency region is a characteristic of solid electrode interface (SEI) formation on the surface of the active material. While the Z_{real} axis with an arc in medium frequency region was charge transfer resistance between electrolyte and electrode interface and Z_{real} axis at low-frequency a region in straight line was associated with the Warburg impedance relative to ion diffusion resistance.^[39] EIS circuit and the operating system (Figure 7(a)).^[40] R_i represents the internal resistance of four samples. R_{SEI} represents the SEI resistance of test battery, R_{ct} assign the charge transfer resistance and C_{dl} represents the

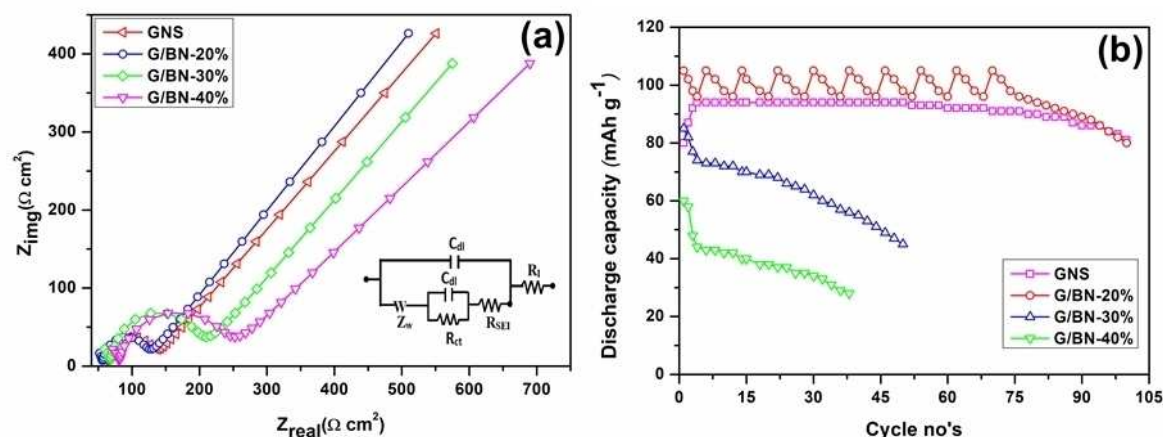


Figure 7. (a) Nyquist plots of pure GNS, G/BN-20%, G/BN-30% and G/BN-40% composites. (b) Rate performance of cycling stabilities for four samples: GNS, G/BN-20%, G/BN-30% and G/BN-40%.

double layer capacitances, and Z_w is denotes the Warburg impedance. The listed fitting parameters obtained from EIS spectra are given in Table 3. Figure 7(a) reveals the G/BN-40% composite electrode with high R_{SEI} of $150 \Omega \text{ cm}^2$ which leads to poor electrochemical performance. The R_{SEI} decreases to $135 \Omega \text{ cm}^2$ in G/BN-30% composite and further reduction occurred in both pure GNS ($105 \Omega \text{ cm}^2$) and G/BN-20% composite ($85 \Omega \text{ cm}^2$) which depict the good electrochemical performance. The charge transfer resistance (R_{ct}) for pure GNS, G/BN-20%, G/BN-30%, and G/BN-40% composite electrodes are 145, 138, 205 and $250 \Omega \text{ cm}^2$ respectively, which revealed the charge transfer resistance gradually increases with increase in high BN substance. The EIS result stated that the G/BN composite with higher BN substance has poor electrochemical performance and high SEI formation due to a semiconducting property of BN. The cycle stabilities of pure GNS and G/BN composites at a current density of 25 mA/g (Figure 7(b)). Figure 7(b) demonstrated that the G/BN-20% composite exhibit capacity at around 105 mAh/g and cycle stability close to 80 cycles when compared to pure graphene and other composites because of its synergetic effect between graphene and BN. If BN increases gradually the cyclic performance decreases due to the semiconducting property of BN. The reversible capacity of

pure GNS and G/BN-20% composite samples slightly decreases after 80 cycles due to the SEI formation on the surface of the electrode.

Conclusion

In conclusion, the graphene supported /BN nanosheets were developed with different BN ratio by using vacuum filtration method followed by thermal condition maintained at 150°C for 2 h. It was observed that the layered BN were supported on the graphene surface. The effect of different BN substance's on the electrochemical properties were investigated systematically. The G/BN-20% composite electrode as a cathode for magnesium battery and it has shown a good discharge capacity of 105 mAh g^{-1} and high cycle stability with excellent rate capability. Superior electrochemical performance of G/BN-20% composite has ascribed the strong synergetic effect between BN and graphene nanosheet. The prepared GNS and G/BN composite materials were used for future research work on energy-related applications such as supercapacitor and battery field like Al/Na storage systems.

Supporting Information Summary

The supporting information included the complete experimental details. Moreover, it can also described the relevant performance of others' work for rechargeable magnesium storage system should be listed in a table 4.

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Electrode	R_i ($\Omega \text{ cm}^2$)	R_{SEI} ($\Omega \text{ cm}^2$)	C_{dl} (F)	R_{ct} ($\Omega \text{ cm}^2$)	C_{dl} (F)
Pure GNS	56	105	0.728×10^{-9}	145	0.292×10^{-6}
G/BN-20%	54	85	0.853×10^{-9}	138	0.309×10^{-6}
G/BN-30%	67	135	0.811×10^{-9}	205	0.321×10^{-6}
G/BN-40%	70	150	0.587×10^{-9}	250	0.322×10^{-6}

Keywords: Boron nitride nanosheets • Graphene • Ionic liquid • Magnesium battery

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